

Muscle Soreness (DOMS) and Injury Prevention

Reference notes distinguishing normal exercise soreness from injury, and general injury-prevention principles. Not medical advice.

What DOMS is

Delayed Onset Muscle Soreness (DOMS) is the diffuse, dull soreness that typically appears 12-24 hours after unfamiliar or unusually intense exercise, peaks around 24-72 hours, and resolves within about 3-5 days. It is a normal adaptive response, most common after eccentric-heavy movements (downhill running, heavy lowering phases of lifts) or a jump in training volume or intensity.

How DOMS differs from an injury

- DOMS: diffuse soreness across a muscle belly, gradual onset over a day, eases with light movement, resolves within about a week.
- Possible injury: sharp, sudden, or localized pain (especially near a joint), pain that appears immediately during the activity, swelling or bruising, reduced range of motion, or pain that does not improve or worsens after several days.

General injury-prevention principles

- Progress training volume or intensity gradually rather than in large sudden jumps.
- Include a warm-up that raises tissue temperature and prepares the specific movement patterns to be used.
- Balance hard training days with easier days to allow tissue adaptation.
- Address persistent asymmetries or nagging discomfort early rather than training through them.

Guidance for the assistant

When a user describes soreness that fits the DOMS pattern (diffuse, gradual, improving with light movement, within the first few days), it is reasonable to suggest light activity or active recovery. When a description matches the injury pattern (sharp, localized, sudden onset, swelling, or not improving after about a week), the assistant should recommend rest from the affected area and, if pain is significant or persistent, seeing a healthcare provider rather than offering a workout modification.